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B. E. (Third Semester) Examination, Nov.-Dec. 2015

(Old Scheme)

(AEI, CSE, EEE, EI, ET & T, IT & Mechatronics Branch)

BASIC ELECTRONICS

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

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Note : *Attempt all questions. Part (a) from each question is compulsory. Attempt any* ***two*** *parts from part (b), (c) and (d) of each question.*

1. (a) Define diffusion capacitance of a diode. 2

(b) Explain in detail how depletion region is formed in a pn-junction. What is its importance for an unbiased

pn-junction. Sketch the variation of space-charge, barrier energy, mobile and immobile carriers? Explain the junction in detail. 7

(c) What do you mean by diode capacitance? Derive expression for transition capacitance. 7

$$CT = \frac{\in A}{W}$$

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(d) The resistivities of the two sides of a step-graded Silicon junction are 5 Ω cm (p side) and 2·5 Ω cm (n side). Calculate the height of the potential barrier V_0. Take $\mu p = 475\ cm^2/Vs$ and $\mu_n = 1500\ cm^2/Vs$ at the room temperature of 300°K and $n_i = 1{\cdot}45 \times 10^{10}$ atoms/cm^3. 7

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2. (a) Define Ripple factor. Mention the ripple factor for HWR and FWR. 2

(b) Explain construction and working of FW center tap rectifier circuit and give comparison between full wave center tap and Bridge rectifier circuit. Why bridge rectifier is preferred over center tap full wave rectifier? 7

(c) Draw VI characteristics of zener diode with the help of circuit diagram. Explain working of zener diode based voltage regulator. 7

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(d) (i) For the circuit shown what values of V can regulate the voltage correctly. Assume $I_L = 0$ to 4 mA and I_Z has safe values from 1 to 5 mA.

(ii) If I_L is fixed at 50/15 mA (ie, $R_L = 15\,\Omega$ k fixed), then find the safe voltage range of V. 7

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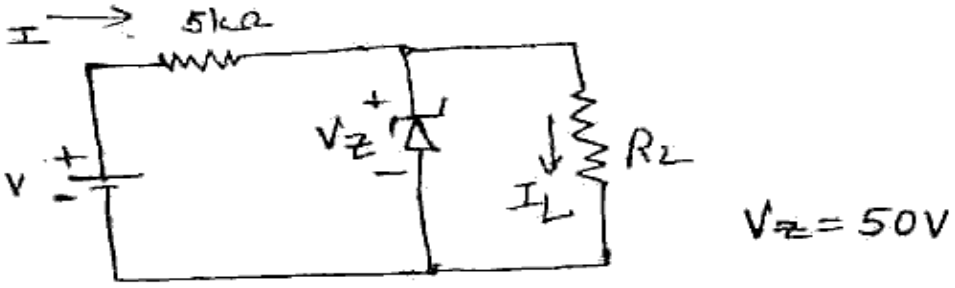

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3. (a) The transistor can act as an amplifier. Justify it. 2

(b) Explain working of npn-transistor in CB configuration. Also explain early effect and three consequences of it with all necessary diagrams. 7

(c) For a certain transistor $I_C = 5{\cdot}255$ mA, $I_B = 100\,\mu A$, $I_{CBO} = 5\,\mu A$. 7

(i) Calculate α, β and I_E.

(ii) Determine new level of I_B required to make $I_C = 15$ mA.

(d) Explain the following terms : 7

(i) Operating point

(ii) Thermal runaway

(iii) Punch through

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(iv) Advantages of CE configuration over CB, CC configuration.

4. (a) What do you mean by Quiescent point? 2

(b) Why NPN type CE amplifier with self bias are mostly preferred in transistor biasing techniques? 7

(c) What is bias compensation and why is it needed? Explain with circuit diagram how is compensation accomplished for variations in base emitter voltage due to temperature variation? 7

(d) Find the Q-point values (I_C and V_{CE}) for the collector feedbadk bias circuit shown in figure. 7

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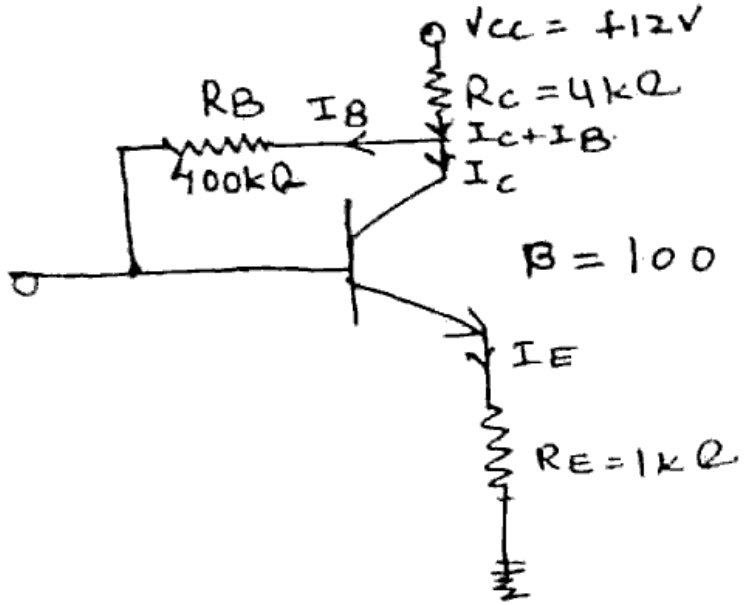

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5. (a) How does the FET behave 2

(i) For small value of V_{DS}, and

(ii) For large value of V_{DS}

(b) Sketch and explain the small signal model of an FET. 7

(i) at low frequencies

(ii) at high frequencies

:) Explain the difference between the enhancement MOSFET and depletion mode MOSFET and sketch the cross-sectional view of depletion type MOSFET.

d) (i) Compare FET with BJT.

(ii) The following information is included on the data sheet for a N-channel JFET.

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$I_{DSS} = 20$ mA, $V_p = -8$ V and $g_{mo} = 5000$ μs

Determine the values of drain current and transconductance at $V_{GS} = -4$ volts.